

Ben Brake

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[Portfolio](#)

Skills

- **Hardware:** Altium Designer (Schematic Capture & PCB Layout), LTSpice, Circuit Design, Soldering, Oscilloscopes, DMMs
- **Mechanical:** Onshape, SolidWorks, AutoCAD, 3D Printing, Machine Shop (Lathe, Mill, Hand Tools)
- **Software:** Python, C++, Java, Git

Experience

- Electrical & Firmware Co-op** – ForceN – Toronto, ON May 2026 – Aug. 2026
- Developed Python-based test fixture, enabling automated testing of HDI PCBA EtherCAT, Ethernet, SPI, and CAN interfaces
 - Designed differentially routed FPC and FFC PCBAs in Altium Designer for internal R&D and external client integration
 - Engineered 4-layer customer-deployed EtherCAT Breakout PCBA using Altium Designer to enable QC and R&D testing
 - Executed board bring-up and validation using precision microsoldering (0201, QFN packages), and coordinated with external mechanical suppliers to deploy test fixtures on the manufacturing floor, maintaining relations and completing on-site rework
- Electrical Project Manager** – Waterloo Aerial Robotics Group – Waterloo, ON Sept. 2026 – Present
- Led schematic and PCB design reviews, mentoring junior electrical members on design accuracy and manufacturability
 - Developed multi-layer PCB stackups and high-current power distribution in Altium Designer, prioritizing large ground planes, low-impedance power routing, and thermal management to support reliable higher-current operation of up to 5-8A
 - Debugged signal integrity, power delivery, and CAN interface faults during bring-up using oscilloscopes and multimeters
- Team Lead** – 4659 Warrior Robotics – Mississauga, ON Sept. 2023 – June 2025
- Led student team to competition finalists at Waterloo VEX Qualifying, leading the design, build, and testing of robots
 - Architected, designed, and integrated custom lifting, shooting, and drivetrain mechanisms, collaboratively iterating through prototyping and testing to meet performance, packaging, and reliability requirements under tight competition deadlines
- Workshop and Sponsorship Coordinator** – WarriorHacks Hackathon – Mississauga, ON Sept. 2023 – May 2024
- Managed event operations for a 100+ participant hackathon, securing sponsorships and coordinating day-of logistics
 - Mentored students in hands-on Python and Java workshops, fostering problem-solving and practical programming literacy

Projects

- 12S -> 5.2V Buck Converter** – [Schematic & PCB PDF](#) Sept. 2026 – Present
- Designed, simulated, assembled, and tested a high current buck converter board using Altium Designer and LTSpice, converting 12S battery input (36V-50.4V) to 5.2V at 8A, enabling safe and reliable power output to fixed-wing UAV servos
- Programming Adapter Rev. 2** – [Schematic PDF](#) Dec. 2025 – April 2026
- Designed intermediary microcontroller programming adapter in Altium Designer, integrating STM32-based USB-to-CAN bridge, STLINK programming interfaces, and configurable I/O to support in-house firmware development and debugging
 - Engineered a configurable 100W USB-C PD system with DC-DC conversion, powering DUTs from a single USB-C source
- Custom Handheld Arcade Emulator** – [Demo Video](#) June 2025 – July 2025
- Engineered a battery powered portable arcade, owning circuit prototyping, 3D printed enclosure modeling, and assembly
 - Configured physical GPIO-driven controls to Raspberry pi emulator inputs to deliver a responsive gaming experience

Education

- University of Waterloo** – BAsC, Mechatronics Engineering Sept. 2025 – April 2030

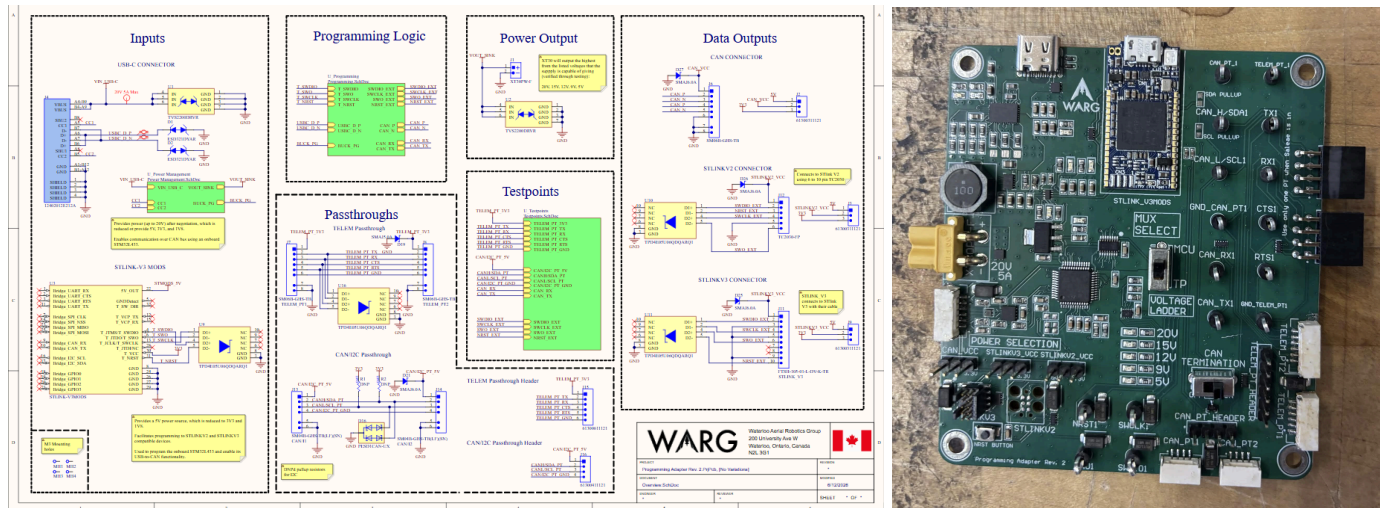
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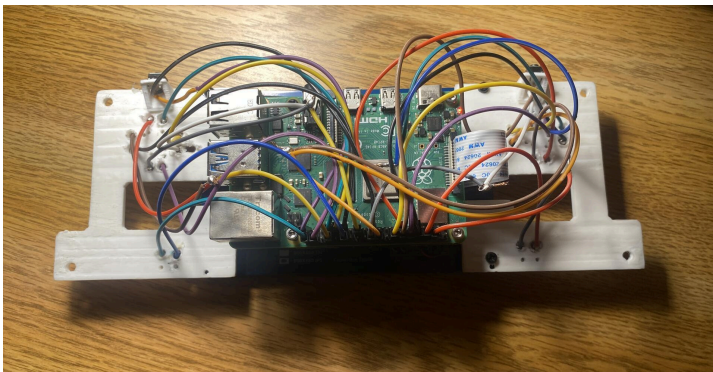
Programming Adapter Rev. 2

- Eliminated custom harnesses during embedded flight systems validation on WARG by consolidating key connectors into one simple programmable interface: [Full Schematic PDF](#)
- Iterated on schematic and layout decisions around STM32L433 integration and STLINK programmability to make the board robust and practical for repeated validation cycles



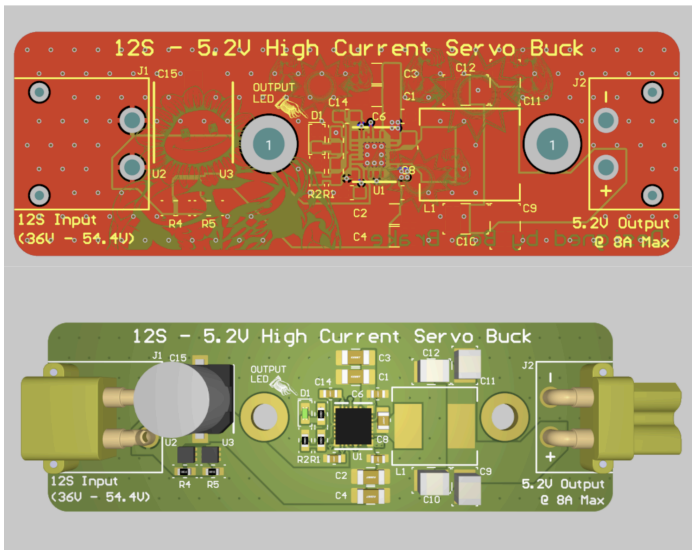
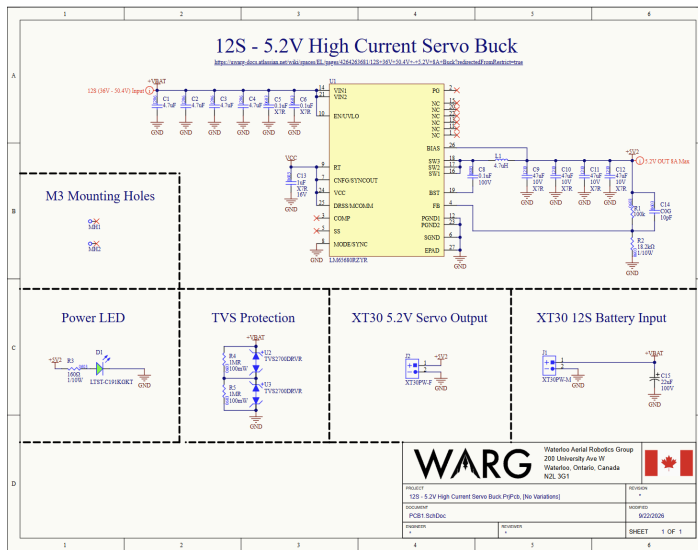
Custom Handheld Arcade Emulator

- Designed and built a fully custom portable arcade unit from the ground up, managing the entire process from circuit design, 3D printed enclosure modelling, to final assembly: [System Demo](#)
- Modeled a custom 3D printed enclosure in Onshape to house all internal electronics in a compact, durable form factor
- Configured RetroPie to map physical controls to emulator inputs, delivering a ready-to-play handheld system
- Prototyped and debugged circuits via breadboarding, then soldered permanent connections for all interface buttons



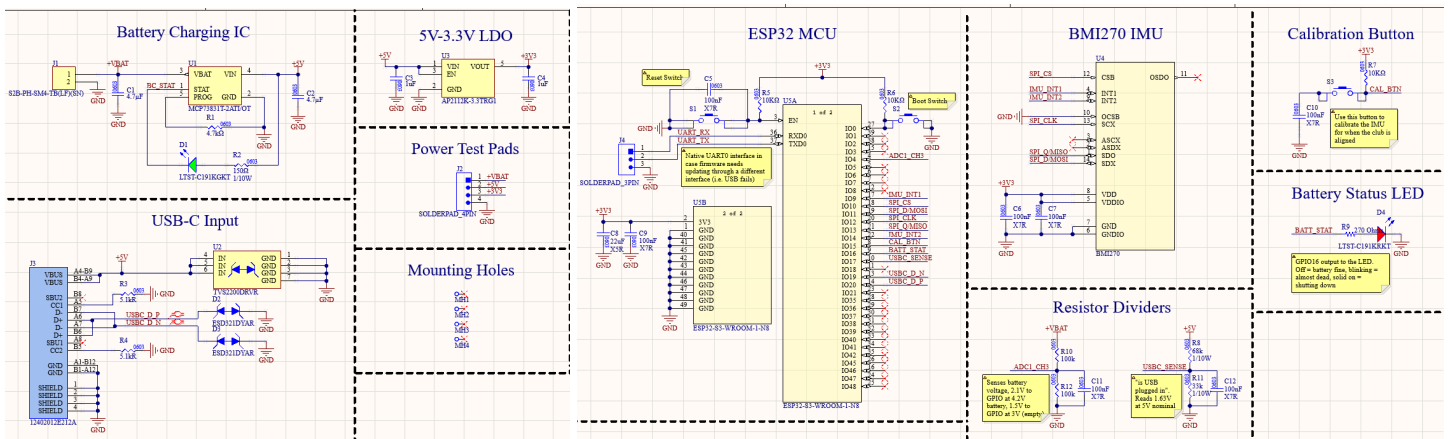
12S -> 5.2V Buck Converter

- Designed for a fixed wing UAV for WARG, this buck converter board steps down the custom 12S battery pack down to 5.2V @ 8A in order to power the wing's aileron 5V servo motors while attempting to account for voltage loss over transmission lines
- With an output of 8A, the LM656x0 series buck converter was used to enable optimization of boards output via simulation



Golf Swing Sensor

- Designed a custom 4-layer PCB in Altium Designer that clips onto a golf club shaft, capturing swing speed, tempo, plane, and face angle via high-rate IMU sensing, BLE module integration
- Developed firmware in MicroPython for event-triggered sensing, and BLE data transmission to a mobile app from the IMU



Autonomous Cornhole Robot

- Designed and built an autonomous robot for MTE 100/121, tasked with automating the lawn game Cornhole: [Project Report](#)
- Using C++ I engineered real-time state machine software to synchronize DC motor actuation with IMU and touch sensor inputs for fully autonomous positioning based on the player's starting position.
- Validated hardware-software integration through iterative bench testing, refining sensor feedback loops and closed-loop motor control for launch consistency

Low-Dropout Regulator Board

- Stepping down 5V to 3.3V, this board was designed in Altium Designer as part of the onboarding for WARG
- Although not the most complicated board, this was a great introduction into the world of PCB design and played a huge role in sparking my love for the field.

